

Electrostatic drainage. Static charge accumulated on the human body and other objects can damage fragile electronic components by discharging through these. Proper grounding in equipment can bypass the path of this discharge current, taking it away from sensitive components. Proper grounding in facilities can reduce damage to sensitive components during manufacturing and handling, too.

EMI control. Grounding/earthing is necessary in EMI control. Shields need to be connected to ground/earth as EMI currents induced in the shield need a path to dissipate to ground. Filters used to reduce conductively-coupled common-mode currents can be effective only if connected to ground to provide a drain for EMI currents. Also, ESD and its associated transient noise is reduced by bonding and earthing equipment frames.

Equipment and system grounding

Grounding hierarchy for a typical installation is shown in Fig. 18. The method of grounding and the type and material of the grounding conductor depend upon the position of the ground in the grounding hierarchy.

At the top of grounding hierarchy (point 1 in Fig. 18) are the low-level low-frequency grounds for circuits that work up to a few kilohertz like audio circuits. A simple wire can be used for such grounds since metals behave as pure resistance at these frequencies.

For low-level high-frequency digital signals up to tens of megahertz (point 2 in Fig. 18), a mesh has to be used for grounding (for example, a bond strap made of wire mesh) since metals start to exhibit inductive properties and meshes are good at providing a low inductance.

For high-level high-frequency EMI currents (which have spec-

tral components up to hundreds of megahertz), grounding has to be done via flat metal strips to a ground plane (point 3 in Fig. 18). This is because metals become highly inductive, and only metal planes and sheets can provide low enough impedance at such high frequencies.

Next in the hierarchy are grounds for cables and transformer shields and those for DC returns of power loads (points 4 and 5 in Fig. 18). Return currents from these should not affect the returns above these in the hierarchy.

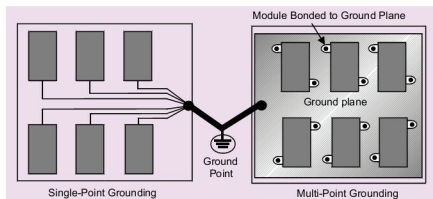


Fig. 19: Hybrid grounding

Lightning and AC power-safety grounds are at the bottom of the hierarchy, and these should also be distinct from one another. Lightning grounds carry hundreds of amperes of current and, hence, the need to separate such grounds from AC power grounds.

Types of grounding

Single-point grounding. Here, each subsystem/module has its own ground. These individual grounds are connected by the shortest route to a single system ground point by simple wires. Such a system is advantageous for low-frequency and analogue circuits because no common impedances exist.

At high frequencies (above 1MHz), however, grounding wires start to exhibit high inductance and, consequently, start offering high

impedance to ground currents. At the same time, capacitive reactance of stray capacitance between the modules starts reducing. Ground currents no longer follow the high-impedance path offered by ground wires to ground, but are rather invited to follow the low impedance offered by the parasitic capacitance to other modules. This causes common-mode coupling, which can be reduced by reducing the inductive reactance of the ground wires. This is achieved by multi-point grounding.

Multi-point grounding. In multi-

point grounding (Fig. 19), each subsystem/module is bonded as directly as possible to a common low-impedance equipotential ground plane (essentially a continuous sheet of metal). A metal sheet offers far less impedance at high frequencies (above 1MHz) compared to wires. This is because sheets have more surface area, which reduces resistance to high-frequency currents that tend to flow on the surface due to skin effect.

The sheet also provides multiple parallel paths for ground currents, which reduce inductance. In such a scenario (when modules are mounted on a ground plane), ground currents are invited to follow the low impedance of the ground plane to ground, rather than going to another module.

Hybrid grounding. Hybrid